Nmap scan types (-sn, -sS, -sT, -sU, -sV, -O, -A) how each works & what it leaves in packets and logs.

1. What is Nmap?

Nmap (Network Mapper) is a network discovery and security auditing tool.

SOC analysts use Nmap to understand network activity, investigate suspicious scans, and identify exposed services.

Penetration testers and administrators also use it for authorized security assessments.
Nmap can identify:

- Live hosts.
- Open ports.
- Closed ports.
- Filtered ports.
- Running services.
- Service versions.
- Operating system fingerprints.
- Network topology information.

Important: Nmap scans should only be conducted on systems and networks for which you have authorization.

2. Understanding the three port states

Port state	Meaning
Open	An application is listening on the port
Closed	The target is reachable, but no application is listening
Filtered	Nmap cannot determine whether the port is open because packet filtering or another network condition prevents a clear response

3. Seven important Nmap scan types

i. -sn - Ping scan / host discovery

Purpose: Discover live hosts without performing a port scan.

Example:

```
nmap -sn 192.168.1.0/24
```

How it works:

- Performs host discovery.
- May use ICMP Echo Requests.

- May use TCP probes, depending on privileges and network conditions.
- On a local Ethernet network, it may use ARP discovery by default.

What it leaves behind:

- ICMP Echo Requests and replies, if ICMP discovery is used.
- ARP requests and replies on local networks.
- Possible TCP probe traffic.

What an analyst may see:

- Many discovery requests from one source.
- Sequential or broad IP address coverage.
- Repeated probes to multiple systems.

SOC interpretation:
A host-discovery scan may indicate network inventory, administration, penetration testing, or reconnaissance.
It is not automatically malicious.

ii. -sS - TCP SYN scan

Also called a half-open scan or stealth scan.

Example:

```
sudo nmap -sS 192.168.1.10
```

How it works:

- Nmap sends a SYN packet.
- If the target replies with SYN-ACK, the port is likely open.
- Nmap generally sends RST instead of completing the connection.
- If the target replies with RST, the port is closed.

Typical packet behaviour:

Target response	**Interpretation**
SYN-ACK	Open
RST	Closed
No response / relevant ICMP error	Filtered or otherwise inconclusive

What it leaves behind:

- TCP SYN packets.
- SYN-ACK responses from open ports.
- RST packets used to terminate half-open attempts.
- Possible firewall or IDS alerts.

SOC interpretation:

Look for:

- Many SYN packets from one source.
- Multiple destination ports.
- Sequential port numbers.
- One source scanning many internal hosts.

iii. -sT - TCP Connect scan

Example:

nmap -sT 192.168.1.10

How it works:

- Uses the operating system's TCP connect functionality.
- Attempts to complete the TCP three-way handshake.
- Establishes a connection to open ports.
- Closes the connection afterward.

Difference from SYN scan:

SYN scan	**Connect scan**
Does not normally complete the handshake	Completes the handshake
Uses SYN/SYN-ACK/RST behaviour	Uses full TCP connection establishment
Often less visible to application logs	More likely to create application-level connection records

What it leaves behind:

- SYN packets.
- SYN-ACK responses.
- Final ACK packets.
- Connection teardown packets, or occasionally resets.
- Potential application and firewall logs.

SOC interpretation:

A connect scan can be easier to identify in application and connection logs because a full TCP connection is established.

iv. -sU - UDP scan

Example:

sudo nmap -sU 192.168.1.10

Purpose: Discover UDP services.

UDP is connectionless, so it does not use the TCP handshake.

How it works:

- Sends UDP probes to target ports.
- If the target returns an ICMP Port Unreachable response, the port is closed.

- If there is no response, the port may be open or filtered.
- If an application responds, the port is generally classified as open.

Typical results:

Response	Interpretation
UDP application response	Open
ICMP Port Unreachable	Closed
No response	Open or filtered
Other relevant ICMP unreachable response	Often filtered

What it leaves behind:

- UDP probes.
- ICMP error messages.
- Application-specific UDP responses.

SOC interpretation:
Look for:

- Large numbers of UDP packets.
- Scanning of DNS (53), SNMP (161), NTP (123), and other UDP services.
- ICMP Port Unreachable responses.

v. -sV - Service version detection

Example:

```
nmap -sV 192.168.1.10
```

Purpose: Identify services and their versions.

How it works:

- First identifies open ports.
- Sends service-specific probes.
- Examines application responses.
- Attempts to identify product names and versions.

Example output:

```
PORT        STATE       SERVICE     VERSION
22/tcp      open          ssh       OpenSSH 9.x
80/tcp      open          http      Apache httpd
443/tcp     open        ssl/http
```

What it leaves behind:

- TCP connections or application probes.
- HTTP requests.
- SSH negotiation traffic.

- Service-specific protocol exchanges.

SOC interpretation:
Version detection is more than simple port scanning because it interacts with the service. Look for unusual protocol requests and repeated probing.

vi. -O - Operating system detection

Example:

```
sudo nmap -O 192.168.1.10
```

Purpose: Estimate the target operating system.

How it works:
Nmap examines TCP/IP stack behaviour, including characteristics such as:

- TCP response patterns.
- TCP window sizes.
- IP ID behaviour.
- TCP options.
- Responses to specially crafted probes.

It compares the results with known OS fingerprints.

What it leaves behind:

- TCP probes.
- Possible ICMP probes.
- Unusual packet combinations.
- Responses with OS-specific network characteristics.

SOC interpretation:
OS detection can indicate reconnaissance, especially when combined with port and service scanning.

vii. -A - Aggressive scan

Example:

```
sudo nmap -A 192.168.1.10
```

This enables several detection features:

- OS detection.
- Service/version detection.
- Default script scanning.
- Traceroute.

What it leaves behind:

- TCP and UDP probes.
- Service-specific requests.

- NSE script traffic.
- ICMP and other traceroute probes.
- Connections to exposed services.

SOC interpretation:
Aggressive scans may generate significant network noise and IDS/IPS alerts.
They are more comprehensive than basic port scans.

4. Nmap comparison table

Scan	Main purpose	Typical network evidence	Possible log evidence
-sn	Host discovery	ICMP, ARP, or TCP discovery probes	Firewall and IDS discovery alerts
-sS	TCP SYN scanning	SYN, SYN-ACK, RST	Firewall, IDS, sometimes connection telemetry
-sT	TCP connection scanning	Full TCP handshake and closure	Firewall, server connection logs, application logs
-sU	UDP scanning	UDP probes and ICMP errors	Firewall and IDS logs
-sV	Service identification	Application-specific probes	Web, SSH, service and IDS logs
-O	OS identification	TCP/IP fingerprinting probes	Firewall and IDS alerts
-A	Aggressive reconnaissance	Combination of probes and scripts	Multiple network and application logs

5. How to detect Nmap activity as a SOC analyst

Consider this example:

A workstation at 192.168.1.50 sends TCP SYN packets to:
192.168.1.100:22
192.168.1.100:80
192.168.1.100:443
192.168.1.100:445
192.168.1.100:3389

Then it repeats the process against multiple IP addresses.

Possible investigation steps:

1. Identify the source IP.
2. Determine whether the source is an authorized vulnerability scanner.
3. Check the number of unique destination ports.
4. Check the number of unique destination hosts.
5. Examine the TCP flags.
6. Look for SYN-ACK and RST responses.

7. Correlate with firewall, IDS and endpoint logs.
8. Determine whether the activity is authorized.

Important SOC distinction: Nmap is a tool, not an attack by itself. An authorized IT administrator can use it legitimately, while an attacker can use the same tool for reconnaissance.